

Networked & Distributed Control: Assignment 1

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Abstract— This report discusses the modeling, controller design, and stability analysis of a sampled-data Networked Control System (NCS) subject to varying sampling intervals and network-induced delays. Both simple and extended state-space models are used to analyze closed-loop behavior under different assumptions. Controllers are designed using state-feedback strategies and evaluated via eigenvalue analysis or LMI-based conditions. The report includes numerical results and stability plots, with a focus on robustness to delays and timing irregularities.

I. INTRODUCTION

This report addresses Assignment 1 of the course SC42101 – Networked and Distributed Control Systems, which focuses on the modeling, analysis, and control of sampled-data systems affected by network-induced delays. The problems covered span material from the first two lectures, and are aimed at deepening understanding of discrete-time control design, delay compensation, and the stability analysis of networked control systems (NCS).

The system under consideration is a linear time-invariant (LTI) plant, where the control signal is implemented in a sample-and-hold fashion and affected by communication delays. The assignment begins with the design of a continuous-time state feedback controller and transitions into the discrete-time domain through exact discretization. Various scenarios are studied, including systems with fixed sampling intervals, systems with constant or varying delays, and finally systems operating under a switching sampling strategy.

For each scenario, appropriate models are derived, controllers are designed, and stability regions are analyzed. The report provides detailed explanations of the modeling steps, control design approaches, and analysis techniques used. Where required, numerical simulations using Python are performed to validate theoretical results and generate stability plots.

The goal of this assignment is to develop the ability to design robust controllers for networked control systems and understand how delays and sampling strategies affect closed-loop stability. All relevant code used to generate results is included for reproducibility.

II. NETWORKED CONTROL SYSTEM

A. Q1: NCS without delays

Fill in $u = -K * x$, which results in our original state space being described by:

$$\dot{\xi}(t) = A\xi(t) + Bv(t)$$

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With:

$$v(t) = -\bar{K}\xi(t)$$

The closed loop poles will be set at $-2 \pm j$, which means the eigenvalues must equal:

$$(s - (-2 - j))(s - (-2 + j)) = 0$$

Rewriting gives:

$$s^2 + 4s + 5 = 0$$

The eigenvalues of our actual system when using values k_1 and k_2 and applying $\det((A - K * B) - s * I)$ is represented by:

$$\begin{vmatrix} k_1 + s & k_2 + 0.5 \\ 2.8 & s + 1 \end{vmatrix} = 0$$

This results in:

$$\begin{aligned} &(k_1 + s) * (s + 1) - 2.8 * (k_2 + 0.5) \\ &= s^2 + s(k_1 + 1) + (k_1 - 2.8 * k_2 - 1.4) \end{aligned}$$

giving $k_1 = 3$ and $k_2 = -3.4/2.8$.

Next, the state space will be discretized using Zero-Order Hold (ZOH). This gives the following state space representation and solution:

$$\xi(s_{k+1}) = e^{A(h)}\xi(s_k) + \int_{s_k}^{s_{k+1}} e^{A(s_{k+1}-s)}Bu(s)ds$$

$$v(t) = u_{k-1} \quad \text{for } t \in [s_k, s_k + \tau)$$

$$v(t) = u_k \quad \text{for } t \in [s_k + \tau, s_{k+1})$$

Using $x_k = \xi(s_k)$, splitting the integral and using a change of variables to define F_x, F_u and G_1 as the individual leftover parts results in:

$$x_{k+1} = F_x * x_k + F_u * u_{k-1} + G_1 * u_k, \quad u_k = -K * x_k$$

where:

$$F_x(h) = e^{Ah}$$

$$F_u(h, \tau) = \int_{h-\tau}^h e^{As}Bds$$

$$G_1(h, \tau) = \int_0^{h-\tau} e^{As}Bds$$

Then an extended state variable x_k^e can be set up which includes the state variable x_k and the input variable u_k :

$$x_{k+1}^e = F * x_k^e + G * u_k$$

Where:

$$F = \begin{bmatrix} F_x & F_u \\ 0 & 0 \end{bmatrix}$$

$$G = \begin{bmatrix} G_1 \\ I \end{bmatrix}$$

However, since there is no delay, this results in F_u being equal to zero. Also to keep dimensions consistent we pad extra zeros, this will be further explained when required for verifying stability in Q4.

To verify stability of this controller we simply have to calculate the spectral radius and find the biggest eigenvalue, if this is smaller then one it is stable, otherwise it is not a stable system. The maximum value of the spectral radii are shown in figure 1.

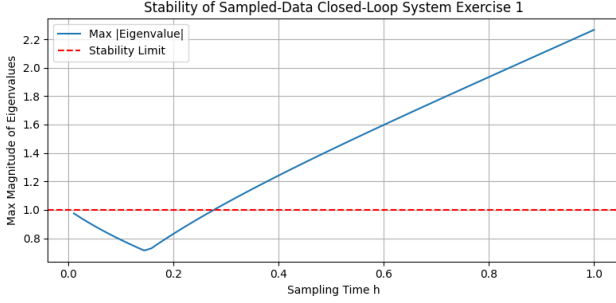


Fig. 1. Stability of the closed loop system with varying sampling time h and τ equal to 0.

Where the highest limit for the sampling time is given by $h = 0.27756756756756756$ and a lower limit that will approach 0.

B. Q2: NCS with constant delays

Next we consider a controller with a constant delay τ . This is a constant value over the same controller mentioned in section II-A. This same model will be used, but with τ non-zero. The values of τ and h are swept over and plotted in figure 2, but since the delay can not be bigger then the sampling time a dotted white line has been added. Red lines enclose the stable area. This results in the actual stable area being the enclosed area above the white dotted line.

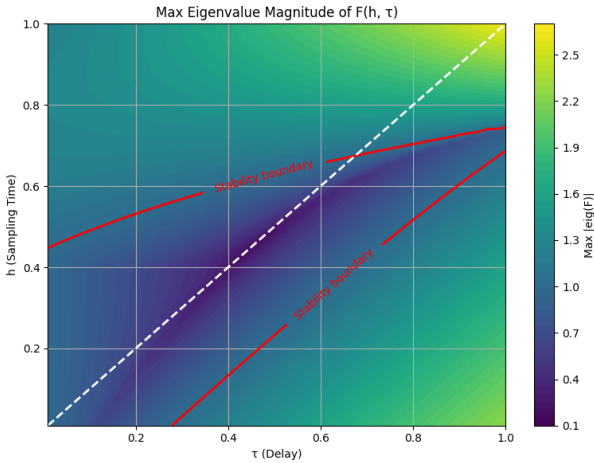


Fig. 2. Stability of the closed loop system given by its spectral radius, with varying sampling time (h) and varying known delay (τ). The red lines enclose the stable region, while the white dotted line defines the line τ equal to h .

To increase the performance of this controller for a stable sampling time, The logical step is to change the values for K . $h = 0.4$ was chosen by observation of the plot, where the optimal K is found by counting the amount of stable systems with a constant τ and sweeping different values of vector K . This results in the following best result of the vector K , where all combinations of k_1 and k_2 are tried in steps of 1 between -10 and 10:

$$K = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}$$

C. Q3: NCS with delays and dynamic controller

This time the system and its state space representation will change slightly since our controller now uses previous values, thus making it a dynamic controller. This causes the extended state space to no longer work.

$$u_{k-1} = -K * x_{k-1} - K_2 * x_{k-2}$$

In our previous extended system we could write u_k in terms of x_k . This is no longer possible and would require an additional previous inputs instead. This results in the following extended state dynamic:

$$x_{k-1}^e = F * x_k^e + G * u_k^e$$

Where:

$$F = \begin{bmatrix} F_x & F_u \\ I & 0 \end{bmatrix} \in \mathbb{R}^{4 \times 4}$$

$$G = \begin{bmatrix} G_1 \\ 0 \end{bmatrix} \in \mathbb{R}^{4 \times 2}$$

$$x_k^e = \begin{bmatrix} x_k \\ u_{k-1} \end{bmatrix} = \begin{bmatrix} x_k \\ x_{k-1} \end{bmatrix} \quad \& \quad u_k^e = [u_k]$$

To evaluate the stability, the same spectral density method will be used as mentioned in section II-A. The results are shown in figure 3. With the red line enclosing the stable area and the white dotted line indicating the τ bigger than h limit.

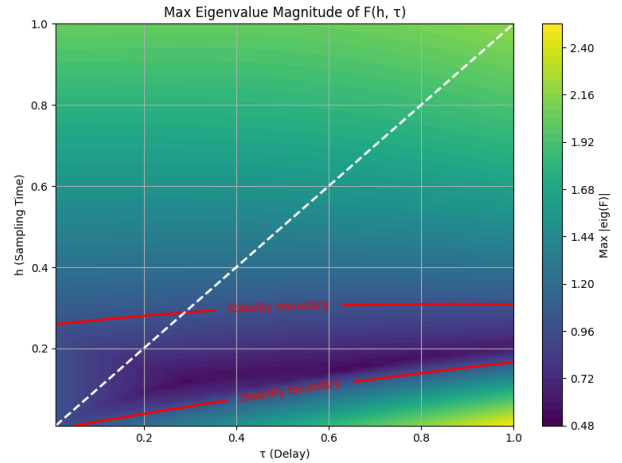


Fig. 3. Stability of the closed loop system given by its spectral radius, with varying sampling time (h) and varying known delay (τ). The red lines enclose the stable region, while the white dotted line defines the line τ equal to h .

To improve the range of valid values for τ using a static sampling we can adjust the values for K again as mentioned in II-B. For this system, This results in the following best result of the vector K , where all combinations of $K1$ and $K2$ vectors are tried in steps of 1 between -10 and 10:

$$K = \begin{bmatrix} K1 \\ K2 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

D. Q4: NCS with varying sample time, no delays and a dynamic controller

Now with τ equal to zero and a switched system combining $h1$ and $h2$ it means eigenvalues no longer guarantee stability, instead a common lyapunov function needs to be searched for. This can be done by finding a valid value for P and Q in a set of LMI's:

$$(F - GK)^T * P * (F - GK) - P < -Q$$

With two sampling intervals h_1 and h_2 , and no delays, we model the system as a switched system. A convex optimization (using cvx) is used with constraints added on P and Q . If for any P and Q bigger then 0 a solution is found for an LMI it means this system is stable and will be added to the plot.

$$P > 0, Q > 0$$

Furthermore the question asks for the optimization to include $h1$ and $h1*h2$. So both of these are added into the constraints, meaning that the stable points for $h1$ and $h2$ will be stable for the system including only $h1$ and the system including $h1*h2$. Then the possible values are shown in figure 4.

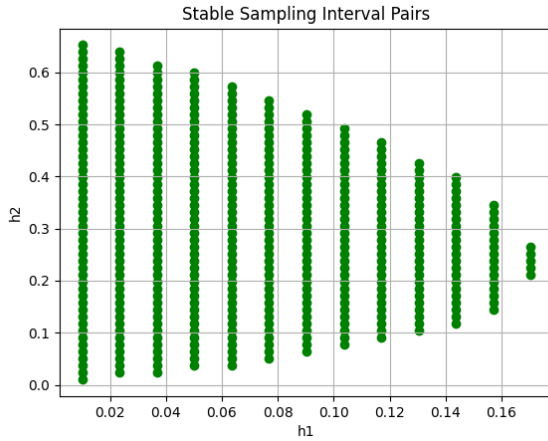


Fig. 4. Stable points for $h1$ and $h2$ following the set of LMI's

III. DISCUSSION

Regarding the difference between Q2 and Q3, where Q3 uses a more complex system. It clearly shows the plot of Q2 has more possible values for h , but suffers by only have a slight range of τ that remain stable. Instead in Q3 it has many more possible values for τ . When taking into consideration that τ

needs to be smaller then h it shows Q2 actually has a better result overall, allowing for more stable h values compared to Q3.

IV. CONCLUSION

This assignment demonstrated how continuous-time controllers can be translated into the sampled domain and how delays and timing uncertainties affect stability. By modeling with extended states and leveraging LMI-based methods, we can design robust controllers suitable for networked environments. Future work could explore stochastic delay models or adaptive sampling strategies.

REFERENCES